



Diversity analysis of methanogens in rumen of *Bubalus bubalis* by 16S riboprinting and sequence analysis

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ABSTRACT

The molecular diversity of rumen methanogens was investigated by 16S rDNA gene library prepared from the rumen contents obtained from Murrah buffaloes in India. Genomic DNA was isolated from adult male fistulated buffaloes and PCR conditions were set up using specific primers. Amplified product was cloned into a suitable vector, and the positive clones were selected assuming based on blue–white screening and sequenced. Positive clones were reamplified and the resulting PCR products were further subjected to Amplified Ribosomal DNA Restriction Analysis (ARDRA) by using *HaeIII* enzyme. A total of 108 clones were examined, and the analysis revealed 16 phylotypes. Out of sixteen phylotypes, nine phylotypes belong to the uncultured group of methanogens, and the rest of seven phylotypes belong to the order *Methanomicrobiales*, *Methanococcales* and *Methanobacteriales*. Out of the 108 rDNA clones, 66 clones which constitute 61.1% of the total clone representing 9 phylotypes, show less than 97% sequence similarity with any of the cultured strain of methanogens. The second largest group of clones (24 clones) represented by four phylotypes show a sequence similarity ranging from 91% to 99% with *Methanomicrobium mobile* strain of methanogens. The third group of 16S rDNA clones clustered along with *M. burtonii* strain of methanogens. This group consists of 6 clones and constitutes about 5.5% of the total clones and represented by only single phylotype. Fourth and fifth clusters of 16S rDNA clones consist of 5 and 7 clones respectively, and these were matched with *Methanobrevibacter gottschalkii* and *Methanobrevibacter rumanatium* strain of methanogens and constitute about 4.6% and 6.4% of the total clones.

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1. Introduction

Enteric methane is produced by methanogenic archaea as a by-product of fermentation of feed in the rumen gut. Reducing the number or activity of these methanogens has been identified as one way of reducing global methane emission inventory (Wright et al., 2007). In order to reduce the activity of methanogens the proper knowledge of different types of methanogens presents in the rumen is essential. There are millions of microbes present in the rumen, including bacteria, protozoa, methanogens and fungi, etc., which will work together in order to supply energy and other nutrients to the host in various forms. Recently, much of interest is generated towards the study of rumen methanogens due to their role in global warming (Wright et al., 2008). It has been estimated that worldwide production of enteric methane from ruminants is around 86 million tons and 18 million tons from manure (Steinfeld et al., 2006).

Methane production is largely affected by the type of feed intake and nature of nutrients present in various feedstuffs, mainly carbohydrates. Different workers generated the information on the changes in diversity of methanogens and on a wide range of bacterial species present in the rumen due to dietary modifications (Kocherginskaya et al., 2001; Marc F Whitford, 2001; Ramšak et al., 2000; Tajima et al., 1999, 2000). The most common species of methanogens isolated and identified from the rumen are strains of *Methanobrevibacter*, *Methanomicrobium*, *Methanobacterium*, and *Methanosarcina* (Jarvis et al., 2000; Steinfeld et al., 2006; Wolin et al., 1997). Methanogens are often found to be associated with the ciliated protozoal population (Finlay et al., 1994; Lloyd et al., 1994, 1996; Tokura et al., 1999a, 1999b) whereas some of the methanogens remains free living bodies. Earlier, methanogens in the rumen have been studied by using classical conventional microbiological techniques (Wolin et al., 1997), but now, with the development of more advanced and specific molecular techniques, their use in studying the diversity of methanogens have been increased (Tajima et al., 1999, 2001; Tokura et al., 1999a, 1999b; Whitford et al., 1998; Yanagita et al., 2000). The present study was undertaken to study the molecular diversity of methanogens present in the rumen of buffalo (*Bubalus bubalis*) using 16S rDNA riboprinting and sequence analysis of different clones.

Abbreviations: 16S rDNA, 16S Ribosomal DNA; ARDRA, Amplified Ribosomal DNA Restriction Analysis; PCR, Polymerase Chain Reaction; NFBRSRA, National Fund for Basic and Strategic Research in Agriculture; ICAR, Indian Council of Agricultural Research.

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2. Materials and methods

2.1. Sample collection and processing

Rumen liquor sample was taken from male fistulated animals. Representative sample of rumen liquor was obtained after manual mixing of rumen contents from five rumen fistulated mature buffaloes (*Bubalus bubalis*) 1 h before morning feed. The buffaloes were kept on a standard diet comprising to concentrate and roughage (wheat straw) in a ration of 40:60. The samples were homogenized using blender and squeezed through two layers of muslin cloth and immediately processed for DNA extraction.

2.2. DNA extraction, PCR amplification, and clone library construction

For DNA extraction, 1 ml of rumen fluid was taken and was processed by using Genomic DNA purification Kit (Fermentas, USA) according to the manufacturer's instructions. All the DNA extractions were performed on fresh samples and the DNA obtained was further stored in deep fridge for downstream processing. PCR amplification was carried out in a Bio-Rad Thermal Cycler by using Met86F and Met1340R set of primers (Wright and Pimm, 2003). The steps in PCR include 95 °C for 5 min, 40 cycles of 94 °C for 1 min, 58 °C for 30 s, 72 °C for 1.5 min and a final extension step of 10 min at 72 °C. A 16S rDNA gene library was constructed from the pooled PCR products obtained from each animal. Cloning of the 1.2 kb PCR product, selection of positive clones on the basis of blue–white screening, and reamplification of the plasmid DNA products were done by standard protocol used in our lab (Chaudhary and Sirohi, 2009).

2.3. Amplified ribosomal DNA restriction analysis

Positive PCR products obtained after the amplification of plasmid DNA that was performed to check the presence of inserts were gel purified by using Wizard SV Gel and PCR Clean-Up System (Promega, USA) and the PCR products obtained after gel extractions were subjected to Amplified ribosomal DNA restriction analysis (ARDRA) by using *HaeIII* restriction endonuclease (New England Biolabs, UK) according to manufacturer's instructions to observe the number of phylotypes present in our clone library. Different phylotypes obtained after ARDRA analysis was sent for sequencing to Bangalore Genei Pvt. Ltd (India).

2.4. Blast and Clustal W analysis

Raw sequences that we obtained after sequencing were aligned by using CLUSTAL W and the resulting sequences were Basic Local Alignment Search Tool (BLAST) to search for the homology between the nearest methanogen sequences. Sequences obtained were submitted to GenBank database with an accession number from GU797091 to GU797106.

2.5. Phylogenetic analysis

Highest similarity sequences in the database entries which were assessed by using an on-line blast search were used as reference in the phylogenetic tree (Madden et al., 1996). Phylogenetic trees were generated based on the neighbor joining algorithm using the MEGA package (Saitou and Nei, 1987; Tamura and Kumar, 2004). Branch stability was assessed by bootstrap analysis (500 replicates) using the algorithms available in the MEGA package (Felsenstein, 1985; Tamura et al., 2007).

2.6. Statistical analysis

Clones with more than 97% sequence similarities were considered as belong to the same species. Biodiversity indices were calculated for methanogens population using the Shannon–Weaver index (H_{Shannon}).

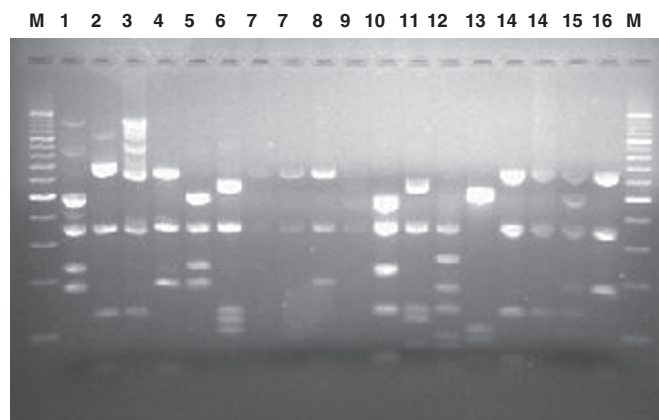


Fig. 1. Lane 1 to 18 showing sixteen different phylotypes obtained during ARDRA profiling of 108 16S rDNA clones. Lane 1 and 20 showing 100 bp Marker.

Index value was used to characterize the methanogenic diversity in the buffalo rumen. Rarefaction analysis was performed using an online tool known as FASTGROUPII (http://biome.sdsu.edu/fastgroup/cal_tools.htm) developed by (Yu et al., 2006).

$$H_{\text{Shannon}} = - \sum_{i=1}^{S_{\text{obs}}} \frac{S_i}{N} \ln \frac{S_i}{N}$$

where S_{obs} is observed number of species, S_i is the number of sequences in the i th OUT (Operational Taxonomic Unit), and N is the number of individual's sample.

2.7. Results and discussion

In the present study, diversity of methanogens presents in the rumen of Murrah buffaloes (*Bubalus bubalis*) was characterized by using comparative 16S rDNA gene sequence analysis. A total of 108, 16S rDNA

Table 1

16S rDNA clones from pooled samples of Murrah buffaloes fed on wheat straw based diet.

16 s rDNA gene phylotype	No. of clones	Size (bp)	GenBank Accession no.	Nearest valid taxon	% Sequence similarity
NDRI-B1	5	1255	GU797091	<i>Methanomicrobium mobile</i>	99
NDRI-B2	9	1257	GU797092	<i>Methanocorpusculum bavaricum</i>	74
NDRI-B3	10	693	GU797093	<i>Methanocorpusculum bavaricum</i>	70
NDRI-B4	8	1255	GU797094	<i>Methanocorpusculum bavaricum</i>	72
NDRI-B5	3	661	GU797095	<i>Methanocorpusculum bavaricum</i>	76
NDRI-B6	6	1258	GU797096	<i>Methanomicrobium mobile</i>	91
NDRI-B7	9	1258	GU797097	<i>Methanocorpusculum bavaricum</i>	74
NDRI-B8	6	1256	GU797098	<i>Methanococcus vannielii</i>	75
NDRI-B9	5	1262	GU797099	<i>Methanobrevibacter gottschalkii</i>	97
NDRI-B10	7	1258	GU797100	<i>Methanobrevibacter ruminantium</i>	98
NDRI-B11	8	642	GU797101	<i>Methanomicrobium mobile</i>	99
NDRI-B12	4	1257	GU797102	<i>Methanocorpusculum bavaricum</i>	74
NDRI-B13	5	1249	GU797103	<i>Methanomicrobium mobile</i>	99
NDRI-B14	9	1254	GU797104	<i>Methanocorpusculum bavaricum</i>	73
NDRI-B15	7	1250	GU797105	<i>Methanocorpusculum bavaricum</i>	72
NDRI-B16	7	1222	GU797106	<i>Methanocorpusculum bavaricum</i>	72

gene clones library of nearly full length were generated using the rumen liquor sample of the Murrah buffaloes (*Bubalus bubalis*) fed on wheat straw based diet. These clones were classified into 16 phylotypes after getting the ARDRA patterns (Fig. 1) and sequencing results (Whitby et al., 2004; Wright and Pimm, 2003). Details of the different phylotypes and similarity percentage with nearest taxon, etc. are presented in Table 1. Out of the 108 rDNA clones, the 66 clones which constitute 61.1% of the total clones and representing 9 phylotypes, show less than 97% sequence similarity with any of the cultured strain of methanogens and form a separate cluster during the phylogenetic analysis and represents the novel uncultured group of methanogens (Fig. 1). These results are in agreement with Koike et al. (2003). They also reported that sheep fed orchard grass or alfalfa hay diet have an uncultured group of bacteria, which was the dominant component of the bacteria population in the rumen. The second largest group of clones which consists of 24 clones and represented by four phylotypes showed a sequence similarity ranging from 92 to 99% with *Methanomicrobium mobile* strain of methanogens. After phylogenetic analysis, all the four phylotypes clustered along with *Methanomicrobium mobile* strain of methanogens (Fig. 2). In our earlier study related with the diversity of methanogens in the

rumen of Murrah Buffaloes by sequence analysis *Methanomicrobium* phylotype comes out to be the dominant phylotype of culturable methanogens present in Murrah buffaloes reared in northern part of India (Chaudhary and Sirohi, 2009).

Results of this study also support the fact that *Methanomicrobium mobile* might be the dominant cultured group of methanogens present in the rumen of Murrah buffalo generally fed on wheat straw based diet. Recent study carried out on rumen methanogens of Surti buffaloes furthermore supports the fact in which after phylogenetic analysis was done of 23 operational taxonomic units (OTUs) based on unique 16S rDNA sequences, they found 12 OTUs (52.17%) affiliated to Methanomicrobiales order. Results of Surti buffaloes also strengthen our results in Murrah buffaloes that Methanomicrobiales is the dominant order of methanogens present in our climatic and dietary conditions (Singh et al., 2011).

The third group of 16S rDNA clones clustered along with *M. burtonii* strain of methanogens. This group consists of 6 clones constituting about 5.5% of the total clones, and is represented by single phylotype (Fig. 2). To our knowledge in this study for the first time clones resembling with *M. burtonii* strain have been identified in the rumen. Earlier,

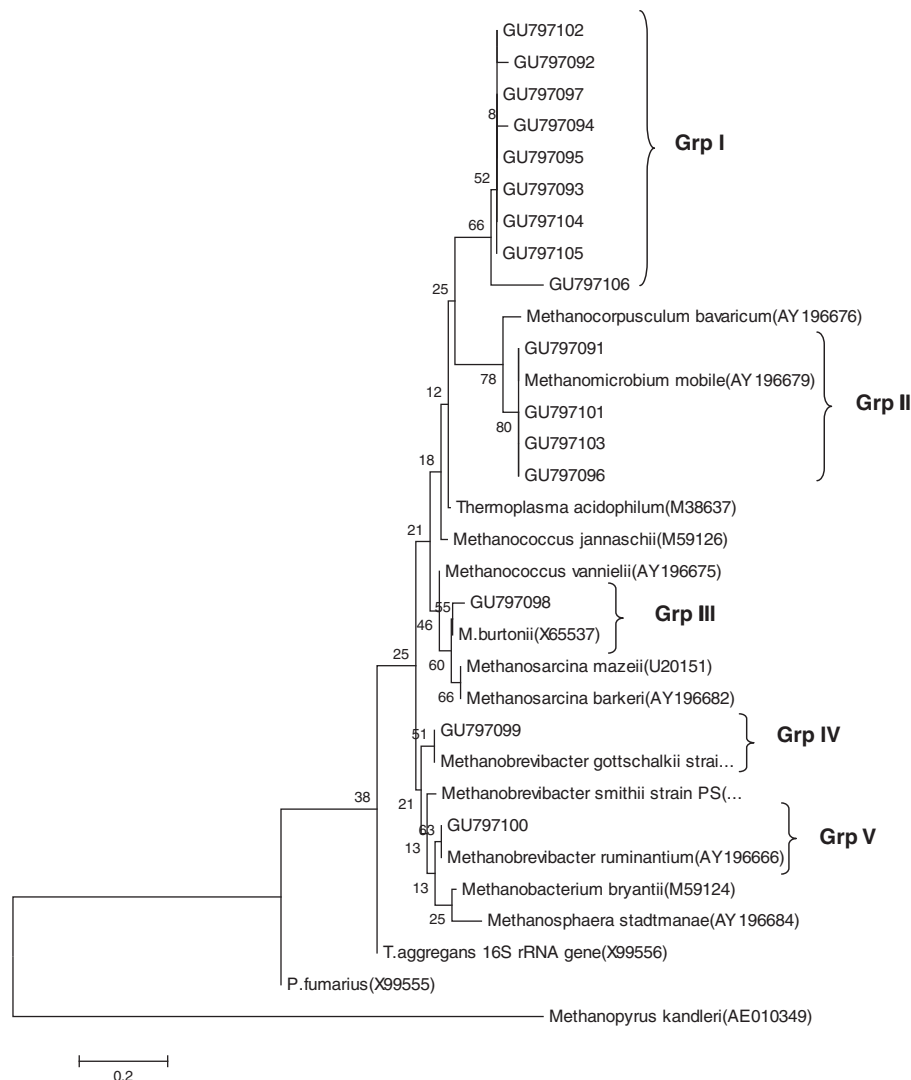


Fig. 2. Evolutionary relationships of 32 taxa. The evolutionary history was inferred using the Neighbor-Joining method. The optimal tree with the sum of branch length = 2.65285048 is shown. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (500 replicates) are shown next to the branches. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Maximum Composite Likelihood method [16] and are in the units of the number of base substitutions per site. Codon positions included were 1st + 2nd + 3rd + Noncoding. All positions containing gaps and missing data were eliminated from the dataset (Complete deletion option). There were a total of 42 positions in the final dataset. Phylogenetic analyses were conducted in MEGA4.

this strain was isolated from Ace Lake in eastern Antarctica (Karr et al., 2006). Fourth cluster of 16S rDNA clones consisting of 5 clones representing single phylotype showed a sequence similarity of 97% and clustered along with *Methanobrevibacter gottschalkii* (Fig. 2). This group constitutes about 4.6% of the total clones under present study. On the contrary in a recent study conducted at Venezuela on sheep, the largest group of clones (41 clones) was 97.9–98.5% similar to *Methanobrevibacter gottschalkii* (Wright et al., 2007). Fifth and last cluster belongs to the *Methanobrevibacter rumanatium* strain of methanogens and 7 clones constituting about 6.4% of the clones library and represented by a single phylotype (Fig. 2). These results were also in contrast with earlier studies carried out in Canada and Australia in which *Methanobrevibacter* phylotype represents the major dominant phylotype of rumen methanogens present in cattle and sheep. In other, study conducted in Canada, 34 clones out of 104 clones were 98.1–99.0% similar to *Methanobrevibacter ruminantium* (Wright et al., 2007). In an earlier study conducted on methanogen diversity in feedlot cattle from Ontario and Prince Edward Island, Canada reported that out of total of 241 clones examined, *Methanobrevibacter ruminantium* accounted for more than one-third (85 clones) of the clones (Wright et al., 2007). This shift in the dominance of methanogen phylotype from *Methanobrevibacter* to *Methanomicrobium* might be due to change in climatic conditions or adaptation of these animals on different dietary conditions in different parts of the world.

Rarefaction analysis was also done to evaluate whether screening of 108 clones was sufficient to estimate methanogen diversity within the prepared clone library or not. All different ARDRA patterns were plotted versus the number of 16S rDNA clones in the clone library. Numerical analyses of the biodiversity provide interesting insight. The mean Shannon–Weaver index is 2.0831. Rarefaction analysis shows that the diversity was almost fully covered in the present study as seen from the curve reaching to plateau (Fig. 3). Community composition of methanogens present in the rumen of Murrah buffaloes was depicted in Fig. 4.

On the basis of the rumen methanogen diversity in the present study, we can say that this study provides useful insight on the future research work in Murrah buffaloes in Northern India. The present study results identified the major type of methanogens inhabiting in the rumen of Murrah buffaloes in India and this information adds to our understanding of community structure of methanogens in this particular breed of buffalo. It also indicated that further studies are warranted in order to properly assess the effect of change in the dietary and climatic conditions, whether influenced by the community

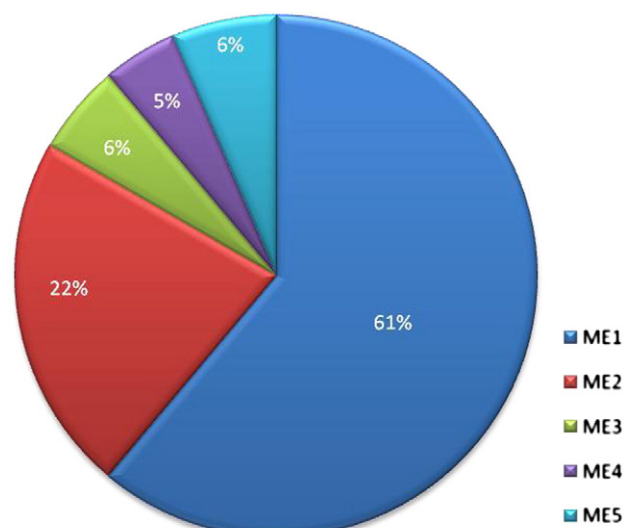


Fig. 4. Spatial distribution of 16S rDNA clones in rumen of Murrah Buffaloes (*Bubalus bubalis*).

structure of methanogens in the rumen or not. Further, as earlier and present study showed the presence of an uncultured group of methanogens in the majority while assessing the diversity of methanogens in the rumen, whether these uncultured groups represent new taxonomic groups and their role in the rumen needs further research.

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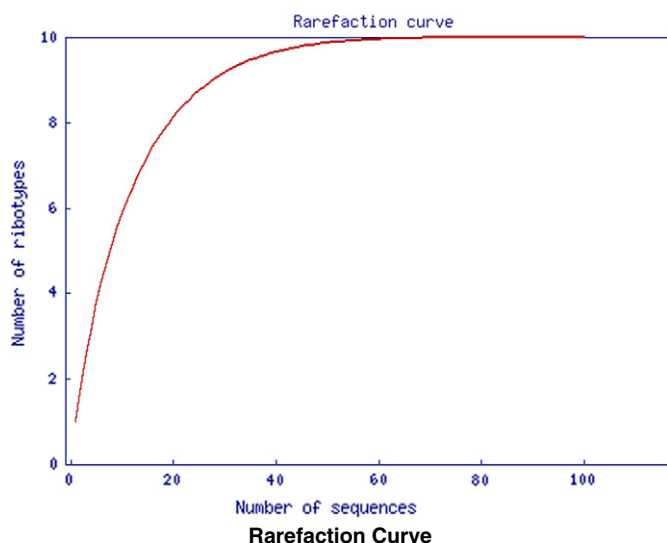


Fig. 3. Rarefaction curve analysis of 16S rDNA clones obtained during the study.

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